METHOD REFERENCE · TCPS-PA v4.0

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Event + Loop · R/A/G/C/L · Dual Contracts · Bidirectional Diagnosis

CARLA 0.9.15 + Apollo 10.0.0 + Bridge

run

Observed / Requirement / Model / Retrospective

2026-08-13

run

“ run ” “ ” run

$$U\_E=(e\_k,B\_b,k,P\_c,k,I\_n,c,k)$$

$$U\_L=(\_q,W\_r,m\_s,\_r)$$

U_E Event-Centric Unit e_k Safety-Relevant Event B_b,k P_c,k source-to-effect path I_n,c,k job/chain instance U_L Loop-Centric Unit _q W_r m_s / _r context “ ” “ hazard ” run

$$T=[R,A,G,C,L]$$

- R Reaction/Response
- A Age/Freshness
- G Fresh Update Gap/Continuity
- C Cross-Source Temporal Coherence
- L Closed-Loop Timing Stability

C “ Age” Fusion/Planning/Control join L “ latency” actuation delay phase drift

L1 L6

L1 Temporal Stressor / Root-Cause		
L2 Local Temporal Degradation		task edge sample-and-hold
L3 Cause-Effect / Loop Timing		actuation
L4 Architectural + Physical Contracts		
L5 Physical Budget Loss		
L6 Physical Safety State		

Evidence Domains run L1 L6 pipeline

```
Event/C4/L6 seed -> L3 event lineage -> L2 local symptom -> L1 hypotheses
Loop/C4-A/physical-behavior seed -> L3 loop window -> L2 jitter/delay primitive -> L1 hypotheses
L4 contract state -> L5 physical budget loss -> L6 physical outcome

L1 verified intervention -> L2 -> L3 -> L4 -> L5 -> L6 -> attribution
```

1. chain task/edge/actuation segment
2. execution queue activation/phase transport staleness reuse gap overwrite/drop actuation / /
3. architectural contract miss physical contract miss Assurance Extension guarantee loss
4.
5. requirement-constrained deadline-excess distance debt /
6. SUT

1.

1.1 “ ”

“ ”

- /
- task/edge/path/bundle/loop Architectural Temporal Contract
- Physical Dynamic Contract
- miss
-
- **SUT** Apollo/Bridge run “ ” “ ”

latency gap

1.2 Observed Diagnostic Core Assurance Extension

Observed Diagnostic Core run / WCET/WCRT

ssurance/Guarantee Extension WCET/WCRT scheduler/priority blocking queue/communication GPU interference multi-rate/sample-and-hold phase actuation bounds

loop window

7. **Observed architectural correctness**
8. **Observed physical correctness**
9. **nalytical/formal timing guarantee**

10. **Single-run empirical headroom** run

Core observed diagnosis Extension NOT_ACTIVATED_MISSING_QUALIFIED_BOUNDS GUARANTEE_NOT_ESTABLISHED observed findings run max P99 formal guarantee

> run

run

1.3

data/observed		
requirement/qualified		
model/predicted		
retrospective/reconstructed	run	

data/observed UNAVAILABLE

1.4

Apollo Bridge Guardian Bridge Control

> ... -> Planning -> Control -> Bridge -> CARLA vehicle dynamics -> physical response

Guardian Control

---

2. Event-Centric Loop-Centric

2.1 Run

run sensor source effect run E2E latency

-
- source-to-sink path
-
-
-

```
run_id + event_id + dependency_bundle_id + chain_id + chain_instance_id
```

hazard

```
run_id + loop_id + window_id + mode_id + context_id
```

U_E Reaction miss U_L PHYSICAL_CAUSE_EVENT Loop timing anomaly

2.2 Safety-Relevant Event

Safety-Relevant Event “ ”

```
E_k = {
  event_id, event_role=PHYSICAL_CAUSE_EVENT, event_type,
  target_id, scenario_geometry,
  cause_predicate, t_c, t_c_interval, clock_domain,
  state_at_cause, discovery_method, prospective_or_retrospective,
  source_evidence_ids, uncertainty, limitations
}
```

- /
-
- /
-
- /

t_c cause_predicate [t_c^L,t_c^U]

DIAGNOSTIC_SEED / Planning STOP margin local minimum PHYSICAL_CAUSE_EVENT cause Reaction contract NOT_TESTABLE_NO_INDEPENDENT_CAUSE “ effect ” hazard reaction

2.3 deadline

run discovery RETROSPECTIVE_EVENT_DISCOVERY

deadline cut-off

```
event discovery may inspect the future
deadline construction may not use post-cause outcome
```

run t_phys run

2.4 Cause-Effect Chain

DAG source-to-sink path source latency

```
LiDAR -> Fusion -> Prediction -> Planning -> Control -> Bridge -> physical effect
Radar -> Fusion -> Prediction -> Planning -> Control -> Bridge -> physical effect
Localization -> Planning -> Control -> Bridge -> physical effect
Chassis -> Control -> Bridge -> physical effect
```

ECRTS 2023 source/sink DAG source-to-sink cause-effect path Yi deadline sensor-to-actuator path task E2E deadline [ECRTS 2023](#) [Yi et al. 2021](#)

path Fusion/Prediction/Planning AND-join

```
dependency_bundle = {
  dependency_bundle_id, event_id, join_semantics,
  member_chain_ids,
  input_role = REQUIRED | CONTRIBUTING | REDUNDANT | OPTIONAL,
  completeness_rule, effect_predicate
}
```

completeness_rule bundle effect event/effect path

2.5 effect

“ ” effect effect predicate

```
Psi_effect(event_type, target_id, vehicle_state, command_state) = TRUE/FALSE
```

effect

- t_cmd Control
- t_apply Bridge /
- t_phys effect predicate
- effect_hold_rule

effect E2E physical Reaction Time

t_phys event/effect type outcome effect_predicate_spec hysteresis

```
[t_phys^L,t_phys^C,t_phys^U]
```

Effect-threshold sensitivity audit a<-0.3,-0.5,-0.8 m/s^2 hold duration contract verdict BOUNDARY_UNCERTAIN_EFFECT_DEFINITION T_R slack D_response D_debt physical-loss uncertainty

2.6 Loop-Centric Analysis Unit

cause

```
loop_unit = {
  loop_id, controller, plant, feedback_sources, command_sink,
  window_id, window_start, window_end, window_selection_rule,
  mode_id, context_id, operating_state_bounds,
  sample_semantics, hold_semantics, physical_tracking_outputs,
  source_evidence_ids, uncertainty, limitations
}
```

driving mode / jitter control mode target-load

Loop-Centric

$$T\_s[n]=t\_sample[n]-t\_sample[n-1],\ J\_s[n]=T\_s[n]-T\_s^{ref}$$

$$T\_feedback[n]=t\_ctrl\_read[n]-t\_state\_source[n],\ T\_actuation[n]=t\_phys\_effect[n]-t\_cmd\_apply[n]$$

controller execution/activation command update gap trajectory/state reuse sensor-to-controller coherence command-to-feedback phase tracking error steering/acceleration rate oscillation metrics timing anomaly window/mode/context “ ” 80 ms

---

3.

3.1

Bridge/SCB	delay	deadline miss
Apollo	fallback solver	
Trace events	read/start/end/pub / monotonic	/
Trace message context	trace_id parent input/output seq source data timestamp	target outcome
Parsed Cyber record	channel /record receipt header/source timestamp payload reuse	read/start/finish
Localization/Chassis	timestamp	Cyber CARLA ground truth
Collision event	actor impulse	
Actor history/ground truth	CARLA frame/sim	Apollo

Apollo Record Cyber RT channel / record timeline
Apollo 10 Cyber RT Developer Tools

3.2

timestamp clock_domain host timestamp_type unit resolution alignment_method error_bound

- wall_epoch epoch
- monotonic monotonic trace
- record_time / record
- header_timestamp header
- measurement/source_time
- CARLA_sim_time/frame
- Bridge_apply_wall Bridge

11.
12.
13. offset/drift/anchor
14. age
15.

P_CLOCK “ ” t_c t_phys Orin

3.3 phase

“ timestamp ” phase “producer consumer CARLA tick / ”
phase quantization phase offset

- clock_alignment_audit.csv
- phase_audit.csv

phase latency PHASE_EFFECT_HYPOTHESIS phase

3.4

```
DIRECT_OBSERVED
OBSERVED_DERIVED
TRACE_LINEAGE
WEAK_TEMPORAL_ALIGNMENT
INDEPENDENT_REQUIREMENT
INDEPENDENT_CALIBRATED_MODEL
VALIDATED_MODEL
```

```
UNVALIDATED_MODEL
RETROSPECTIVE_RECONSTRUCTION
OUTCOME_TRUNCATED
CONFLICTED_EVIDENCE
MISSING
```

CLOCK_TAINT TARGET_TAINT MODEL_TAINT RETRO_TAINT OUTCOME_TRUNCATED

3.5

- P_TARGET sensor/Fusion/Prediction/Planning/Control/ actor
- P_FUNC Perception Prediction Planning trajectory Control trajectory Bridge

P_FUNC=QUALIFIED_PASS C4_ARCH C4_PHYS “ ” Planning STOP trajectory infeasible Control

P_BUNDLE P_LOOP P_EFFECT

- P_BUNDLE consumer job join role source epoch
- P_LOOP controller/plant/feedback window selection mode/context
- P_EFFECT effect predicate hold filter L/C/U

Coherence Loop physical Reaction claim ceiling P_CLOCK/P_TARGET

3.6

CLOCK_TAINT TARGET_TAINT MODEL_TAINT RETRO_TAINT OUTCOME_TRUNCATED

- BUNDLE_TAINT join consumed instance nearest message
- COHERENCE_COMPENSATION_TAINT epoch /
- WINDOW_SELECTION_TAINT loop window mode/context
- EFFECT_DEFINITION_TAINT effect threshold/hold/filter outcome sensitivity
- RESOURCE_ATTRIBUTION_TAINT residual CPU/GPU/lock/scheduler

taint C3/C4/C5/C7

---

4. L3 Event Chain Loop Timing

run L3/L4 L6 seed loop-window contract physical tracking anomaly L1

4.1

trace record Localization Bridge

```
event_time.csv
{
  event_uid, run_id, event_id, chain_id, module, stage,
  event_kind, read_or_write, clock_domain, host,
  raw_time, aligned_wall_time, alignment_error,
  input_seq, output_seq, trace_id, parent_trace_id,
  data_source_time, target_id, payload_digest,
  source_file, source_locator, evidence_class, quality_flags
}
```

event_kind physical_cause sensor_sample receive/read ready start end publish/write command bridge_apply physical_effect outcome

4.2

Grade		
A	trace_id parent/provenance input/output seq	cause-effect propagation
B	source/header timestamp + downstream mapping	propagation
C		/
D	nearest-time heuristic	
UNKNOWN		

P99 stage decomposition causal chain instance

4.3 Forward causal tracing

source sample “ source ” job

```
source sample -> earliest eligible downstream read -> ... -> first qualifying command -> physical effect
```

- source data sink
- consumer
-
- /
- Control Bridge

4.4 Backward provenance tracing

effect Control Planning

```
physical effect <- applied command <- Control input trajectory
                <- Planning inputs <- Prediction/Fusion source samples
```

“ ” Data Age reuse/staleness Forward backward

- forward sample effect
- observed effect sample
- overwrite skip reuse

4.5 ECRTS partition

ECRTS 2023 p-partitioned job chain backward chain forward chain warm-up MRT MDA partition point

```
at partition p:
backward provenance from p to source
+ forward propagation from p to physical effect
```

-
- read/write job warm-up/valid-chain job-chain
- PARTITION_DIAGNOSTIC_VIEW MRT/MDA

4.6 Multi-input Age Vector

Planning/Control Age

$$A_{c,k}(t_e)= [A_{perception},A_{prediction},A_{localization},A_{chassis},A_{map},\ldots]$$

$$A_j^{read}=t_{read}-t_{source,j},\ A_j^{cmd}=t_{cmd}-t_{source,j},\ A_j^{apply}=t_{apply}-t_{source,j},\ A_j^{phys}=t_{phys}-t_{source,j}$$

Age t_effect/read source semantics chain dependency_bundle_id input_role target relevance clock error A_critical REQUIRED/CONTRIBUTING join/completeness vector critical source

Age join bundle consumption instance q source-time coherence

$$K_B(q)=\max_{j\in B} t_{source,j}-\min_{j\in B} t_{source,j}$$

pairwise skew matrix

$$K_{ij}(q)=t_{source,i}-t_{source,j}$$

topic consumer job bundle role/completeness Fusion measurement time state-estimation epoch prediction/compensation target epoch consumer read time fusion epoch

```
raw_source_skew
compensated_state_epoch_skew
compensation_model/version/horizon
residual_time_alignment_uncertainty
```

timestamp header coherence Temporal Coherence requirement τ_C^{arch} Fusion/Planning $\tau_C^{phys}(x)$ COHERENCE_OBSERVED INCOHERENCE_FAILURE

4.7 Fresh Causal Update Gap

output gap effect q_n

```
qualifying(q_n) iff
  target/event semantics match
  AND source timestamp strictly advances
  AND lineage reaches the selected sink/effect
  AND output is not mere reuse of the same source state
```

$$G^{fresh}_n=t(q_{n+1})-t(q_n)$$

- $G_{publish}$
- $G_{source\_advance}$ source timestamp
- G_{fresh_effect} sink/effect

“Control 100 Hz ” “ 10 ms ” Control Planning trajectory

---

5. L2

5.1 task/edge

producer i-1 consumer i

```
t_write_prev  producer
t_receive     consumer/record
t_ready       job
t_start       job
t_end         job
t_write       consumer
```

$$L_i^{transport}=t_{receive}-t_{write,prev}$$

$$L_i^{queue}=t_{ready}-t_{receive},\ L_i^{activation}=t_{start}-t_{ready}$$

$$L_i^{exec}=t_{end}-t_{start},\ L_i^{publish}=t_{write}-t_{end}$$

ready start-receive queue phase unresolved_wait record
record

5.2

Sampling delay	t_s-t_c	sensor tick phase
Transport delay	receive - producer write	SHM/RTPS bridge
Queue/backlog	ready - receive /	CPU/GPU contention callback backlog
Activation/phase wait	start - ready	timer/callback phase
Execution tail	end - start	fallback
Publish delay	write - end	writer/backpressure
Staleness	consumer time - source time	producer
Reuse	sink source/trajectory	sample-and-hold gap
Overwrite/skip	source update downstream successor	buffer policy
Drop/reorder	seq /	transport/queue policy
Actuation delay	physical effect - applied command	Bridge

5.3 R/A/G/C/L

T_deg=[R,A,G,C,L]

- R reaction/response-time behavior
- A data freshness/age
- G update continuity/gap
- C multi-input bundle source-time skew pairwise skew residual incoherence
- L sample interval/jitter feedback delay actuation delay phase drift reuse/hold

run count P50 P90 P95 P99 MAX IQR/MAD run
run formal bound C bundle/consumer instance L loop/window/mode/context

5.4 run “ ”

L2 “ ” reference run

- NOMINAL_PERIOD
- ENGINEERING_REQUIREMENT /edge timing requirement
- PRE_FAULT_WITHIN_RUN run
- DECLARED_EXTERNAL_REFERENCE
- LOCAL_STATIONARY_REFERENCE run

context covariates defeater

reference

> run 412 ms Planning

> Planning continuity degraded

5.5

Prediction -> Planning 85 ms edge/segment phase queue scheduler interference ready/start / L2

```
observed symptom
candidate mechanisms
supporting evidence
challenging evidence
missing discriminator
```

5.6 Loop-Centric

U_L

Sampling	T_s,J_s missed/extra samples	timer release message arrival controller execution
Feedback	source-to-read age feedback delay phase offset	controller state instance
Controller	activation wait execution publish	ready/start unresolved wait
Hold/reuse	command hold trajectory reuse fresh command gap	
Actuation	command-to-apply apply-to-response deadband/build-up	effect
Physical behavior	tracking error overshoot oscillation rate/slew	outcome/behavior timing cause

Loop anomaly observed Architectural Contract “ / ” delay/jitter margin validated simulation/reachability envelope LOOP_TIMING_ASSOCIATION

L

```
L^obs=[T_s,J_s,T_feedback,T_controller,G_cmd^fresh,T_actuation,_loop,H_reuse]
```

tracking error overshoot oscillation L physical/control behavior vector L composite loop verdict required components NOT_TESTABLE

---

6. L1 Temporal Stressor

6.1

Discovery mode L1 L4 miss L6 loss L3/L2 anomaly L1 hypothesis “ ” stressor entry

Validation mode L1 “ delay” delay

Claim Ledger C1_STRESSOR_ENTRY “ stressor/ ” diagnosis hypothesis ledger C1 OR

6.2 Temporal Fault/Stress Signature

```
F_T = {
  fault_or_stressor_type, location, onset, end, duration,
  requested_magnitude, actual_magnitude, distribution,
  persistent_or_transient, one_shot_or_repeated,
  affected_channels/messages, queue_policy,
  drop_status, reorder_status, resource_context,
  direct_evidence, confidence
}
```

- sampling/trigger delay
- execution/WCET tail
- CPU/GPU/memory/lock contention
- scheduler/priority/blocking
- queue/backlog/backpressure
- network/transport/Bridge delay
- period/phase/mode-transition delay

- staleness/reuse/overwrite/drop/reorder
- actuation/control-build-up delay
- clock/timestamp artifact
- functional or physical alternative

6.3

“ ” applied row Bridge/SCB Bridge/Control Apollo

fault onset t_c [t_f,t_c] pre-hazard state divergence /

6.4 Resource Interference Graph

L2 execution/unresolved-wait L1 Resource Interference Graph “Planning execution tail” task/job CPU runnable/running mutex/futex GPU submit/queue/kernel memory/IO network/SHM logger/filesystem RUNS_ON WAITS_FOR BLOCKED_BY PREEMPTED_BY CONTENDS_WITH SUBMITS_TO

job

$$L_i^{job}=L_i^{CPU\ running}+L_i^{ready\ not\ running}+L_i^{blocked}+L_i^{GPU\ queue}+L_i^{GPU\ kernel}+L_i^{sync}+L_i^{IO}+L_i^{unknown}$$

scheduler trace thread state lock/GPU instrumentation resource counter residual wall execution - CPU time scheduler delay resource_interference_graph.csv resource_interference_edges.csv / job discriminator EXECUTION_SEGMENT_LOCALIZED

---

7. L4 Architectural Physical

L4 “ deadline” U_E U_L L4-A Architectural Temporal Contract L4-B Physical Dynamic Contract

7.1 L4-A Architectural Temporal Contract

Architectural Contract “ ”

- task release/period/deadline execution budget
- producer-consumer edge age/freshness
- module publish/update gap
- source-to-sink E2E reaction/age

- multi-input bundle coherence/source skew
- sample/phase relationship command hold/reuse
- loop sampling jitter feedback delay actuation delay delay/jitter margin

```
architectural_requirement = {
  requirement_id, owner, artifact, version, interface/task/path/bundle/loop scope,
  metric_semantics, value_or_bounds, unit, applicable_mode/context,
  timestamp_basis, aggregation_rule, miss_policy,
  effective_time, lock_time, provenance, qualification
}
```

/ / / > > run
"Apollo 10 Hz" QUALIFIED_ARCHITECTURAL period execution time age

Architectural miss ARCHITECTURAL_TIMING_FAILURE backward diagnosis seed

7.2 L4-B Physical Dynamic Contract deadline

x_c ρ / θ H

$$\Phi(x_c,\rho;\theta)= \min_{t\in[0,H]} h(x_{ego}(t),x_{target}(t),geometry,policy)$$

Φ≥0 dynamics safety buffer deadline

$$\tau_{R,c,k}^{req}= \sup\{\rho\geq 0 \mid \forall r\in[0,\rho], \inf_{\theta\in\Theta_m}\Phi(x_c,r;\theta)\geq 0\}$$

Θ_m context/ ODD " " Φ ρ
safe-delay intervals 0 deadline deadline
x_c

7.3 /

RSS-like

$$d_{req}(\rho)=v_e\rho+\frac{1}{2}a_{resp}\rho^2+\frac{(v_e+a_{resp}\rho)^2}{2b_e}-\frac{v_f^2}{2b_f}+d_{safe}$$

$d_{req}(\rho)\leq d_{clear}$ ρ $v_f=0$ ρ=0

```
tau_req = 0
state = ALREADY_OUTSIDE_DECLARED_ENVELOPE
```

delay

Koopman Osyk Weast longitudinal model response period braking trajectory closest approach Koopman et al. 2019

reachability/trajectory model deadline NOT_QUALIFIED_PRIMARY

7.4 Physical Deadline

deadline observed miss

- state input t_c cut-off
- target identity geometry target-motion assumption
- / dynamics braking envelope
- calibration runs evaluation run
- friction grade curvature actuation build-up uncertainty bounds
- run post-t_c t_phys
- tau_req_low/center/high provenance

UNVALIDATED_MODEL RETROSPECTIVE_RECONSTRUCTION NOT_QUALIFIED_PRIMARY

7.5 ODD / context uncertainty

μ context region m

$$\tau\_robust(m)=inf\_x\in X\_m\tau(x)$$

“ region ” ODD “region ” “ region ” Koopman et al. 2019

7.6

Reaction Freshness Fresh Update Continuity Temporal Coherence Loop Timing architectural physical

$$S\_arch=\tau\_arch^req-[R,A,G,C,L]^obs$$

$$S\_phys=\tau\_phys^req(x)-[R,A,G,C,L]^obs$$

threshold

- Freshness contract “ a / ”
- Fresh-update contract “ g / reachable states ”
- Coherence contract “ skew k / ”
- Loop contract sampling/delay/jitter margin
- reaction deadline nominal period

ECRTS warm-up MRT MDA job-chain observed Reaction Age A/G ECRTS 2023

```
verdict_arch_[R|A|G|C|L] = compare(observed, qualified_arch_requirement)
verdict_phys_[R|A|G|C|L] = compare(observed, qualified_physical_requirement)
```

```
architectural_contract_verdict / physical_contract_verdict:
  MISSED      if any applicable required component is CLEARLY_MISSED
  WITHIN      if every applicable required component is CLEARLY_WITHIN_CONTRACT
              and no required component is missing
  UNCERTAIN   if no component is missed and at least one is BOUNDARY_UNCERTAIN
  NOT_TESTABLE if no component is missed/uncertain and a required component is missing
```

required_components event type dependency bundle safety model NOT_APPLICABLE NOT_TESTABLE

ARCHITECTURAL_MISSED + PHYSICAL_WITHIN “ ” PHYSICAL_MISSED Safety-Critical Timing Failure

7.7 observed contract verdict

[T_L,T_U] deadline [τ_L,τ_U]

T_U≤τ_L	CLEARLY_WITHIN_CONTRACT
T_L>τ_U	CLEARLY_MISSED
	BOUNDARY_UNCERTAIN
deadline	RETROSPECTIVE_ONLY
	MODEL_SUPPORTED_ONLY
/clock/target	NOT_TESTABLE

observed slack

7.8 Observed correctness Assurance/Guarantee Extension

$$S_{obs}=\tau_R^{req}-T_R^{obs}$$

$$S_{guar}=\tau_{R,low}^{req}-R_{path}^{UB}$$

R_path^UB / / WCET/WCRT release/period priority/scheduler blocking queue/transport multi-rate overwrite semantics phase actuation bound

Yi Σ_i(p_i+r_i) 2Σ_i p_i CPU EDF WCET Apollo budget Apollo trace period formal guarantee Yi et al. 2021

S_obs≥0 S_guar<0

R^UB S_guar=NOT_ESTABLISHED

> single_run_empirical_headroom = tau_req - max observed comparable instance

comparable instance event type target semantics dependency bundle/chain mode context region effect predicate retrospective seed selection bias guarantee

Assurance Extension Core bound_assumption_audit=QUALIFIED S_guar guarantee-loss points extension exact bounds verdict

7.9 Dynamic deadline shrink mode transition

Yi deadline deadline job sensor data E2E deadline Yi et al. 2021

hazard window contract trajectory

$$d/dt\ S(t)=d/dt\ \tau^{req}(t)-d/dt\ T^{risk}(t)$$

S>0,d/dt S<0

$$T_{exhaust}^{local}=(S)/(-d/dt\ S)$$

deadline region/mode change job trajectory contract

7.10 Slack Waterfall Guarantee-Loss Point

chain i

$$L_{prefix,i}^{obs}=t_i-t_c$$

$$B_i^{obs}=\tau_R^{req}-L_{prefix,i}^{obs}$$

i effect suffix bound $R_i \longrightarrow phys^{UB}$

$$B_i^{guar}=\tau_{R,low}^{req}-L_{prefix,i}^{obs,U}-R_i \longrightarrow phys^{UB}$$

suffix suffix path/bundle mode phase backlog clock basis / observed prefix suffix wait/exec GUARANTEE_NOT_COMPOSABLE sampling transport queue/unresolved wait activation execution publish consumer wait command-to-apply actuation

16. GUARANTEE_NOT_ESTABLISHED_FROM_START B_0^guar<0 “ ”
17. FIRST_CONDITIONAL_GUARANTEE_LOSS B_i^guar<0 “ ”

18. FIRST_IRREVERSIBLE_OBSERVED_MISS L_prefix,i^obs,L>τ_U effect miss
19. OBSERVED_EFFECT_MISS t_phys miss

observed_contract_miss_time elapsed-time τ_U miss t_phys endpoint/alignment uncertainty FIRST_CONDITIONAL_GUARANTEE_LOSS “ + ”

> causal_interpretation = NOT_INFERRED
> root_cause_status = REQUIRES_BACKWARD_DIAGNOSIS

segment budget/reference

suffix bound observed slack waterfall guarantee-loss point

---

8. seed

8.1 seed

20. L4-B physical contract miss
21. L4-A architectural contract miss
22. L6 L4
23. L3 R/A/G/C loop-window timing/physical behavior
24. L2 reference
25. reference case-level anomaly

seed C1

8.2 Backward diagnosis

seed

26. **seed** chain outcome
27. **lineage/loop segment** effect/command source feedback→controller→command→plant→feedback contract
28. sampling execution queue activation/phase transport staleness/reuse coherence skew overwrite/drop mode transition feedback/actuation resource interference
29. clock artifact target mismatch functional failure initial unsafe state braking/road/geometry outcome conflict
30. onset seed
31. supporting challenging/contradicting missing evidence
32. equivalence class

33. **discriminator** trace /
34. /

8.3

```
SUPPORTED_CANDIDATE
CONSISTENT_BUT_UNRESOLVED
CHALLENGED
REFUTED
NOT_TESTABLE
NOT_APPLICABLE
```

DETECTED	/
LOCALIZED_TO_SEGMENT	
MECHANISM_SUPPORTED	
ISOLATED_CANDIDATE	
INTRINSIC_DEFECT_SUPPORTED	/ /

run “ ” LOCALIZED_TO_SEGMENT ISOLATED_CANDIDATE

8.4 Forward validation

```
verified stressor
-> matching local primitive changes
-> causally matched chain timing effect
-> qualified contract state
-> physical budget loss
-> outcome
```

backward candidate

---

9. L5 Physical Safety Budget Loss

9.1 Canonical observed response distance

cause t_c effect t_phys

D_response,wall^data/observed= ∫_t_c^{t_phys}v(t) dt_wall

```
D_response_wall_integral_data_observed_m
```

D_delay_wall_integral_data_observed_m CARLA frame/sim time Localization realtime factor

9.2 Deadline-excess distance debt

deadline endpoint

```
[t_d^L,t_d^U]=[t_c^L+τ_L,\ t_c^U+τ_U]
```

miss deadline q∈,center,high\

```
D_debt,q^requirement\ constrained= max(0,∫_t_c+τ_q^{t_phys}v(t) dt_wall)
```

D_debt_req_low/center/high_m t_c,t_phys clock alignment wall timestamp longitudinal speed / gap threshold INCOMPLETE_COVERAGE sample count coverage ratio gap “ ” “ ” ancestry “ requirement + observed velocity path” REQUIREMENT_CONSTRAINED_DERIVED

- model deadline -> D_debt_model_predicted_m
- retrospective deadline -> D_debt_retro_diagnostic_m
- deadline -> D_debt

9.3

```
M_0^data/observed=D_1-D_response^obs-D_brake^obs

M_safe^requirement\ constrained=M_0^obs-D_safe^req
```

D_1 D_response D_brake fault onset t_c

- Total Closed-loop Effect t_f outcome
- Post-cause Response Effect t_c

response distance deadline-excess debt

9.4

/

```
C_min^data/observed= min_t∈[t_c,t_o]clearance(x_e(t),x_o(t),geometry)
```

t_o outcome/ x_o target-identity actor measurement source measurement time transport age geometry/shape coverage CARLA ground truth/actor history outcome Apollo “ ”

/ requirement-constrained margin

$$M\_phys^{requirement\backslash constrained} = \min_{t\in[t\_c,t\_o]}[clearance\_obs(t)-d\_safe^{req}(t)]$$

d_safe^req h reachable-set envelope model/predicted observed

$$M\_phys^{model/predicted}(\rho) = \min_t h\_model(x\_e(t;\rho),x\_o(t),\Theta\_m)$$

slack margin margin margin

9.5 “ ”

run

35. D_response_wall_integral_data_observed_m
36. D_debt_requirement_constrained_m deadline
37. minimum_clearance_data_observed_m + contact/collision outcome
38. safety_margin_requirement_constrained_derived_m clearance
39. Delta_M_phys_model_predicted_m ρ_ref

$$\Delta M\_phys^{model} = M\_phys^{model}(\rho\_ref) - M\_phys^{model}(T\_R^{obs})$$

5 “ timing X ”

ρ_ref outcome requirement rho_ref_source rho_ref_lock_time zero-delay deadline-compliant response nominal controller response run ρ_ref

9.6 Collision right-censoring

run

- collision actor time/frame impact speed/impulse
- observed response/braking segment
- actual clearance/contact state

“ observed braking distance full-stop margin” UNAVAILABLE_OUTCOME_TRUNCATED observed

---

10. L6 Physical Safety State

L6

- minimum/final clearance
- physical/contact/engineering margin
- minimum speed strict stop near stop
- impact speed impulse collision actor

- truncated braking time/distance
- safety envelope violation duration/area

outcome

```
SAFE
SAFE_LOW_MARGIN
CRITICAL_MARGIN
MINIMUM_DISTANCE_VIOLATION
EMERGENCY_MANEUVER
NEAR_MISS
CONTACT
COLLISION
HIGH_SEVERITY_COLLISION
UNCERTAIN
```

CONTACT/COLLISION low/critical/near/high-severity provenance

OUTCOME_OBSERVED MARGIN_CLASSIFICATION outcome " / " requirement/reference run clearance degraded

/

---

11. run

11.1 Core Profile run

Core Profile Observed Diagnostic Core / event catalog loop catalog lineage R/A/G/C/L primitives Architectural/Physical contract qualification effect sensitivity outcome Claim–Evidence summary

11.2 Extended Profile

Extended Profile / contract miss outcome loop anomaly execution segment qualified path/suffix bound / Extended full defeater ledger Resource Interference Graph formal/suffix bounds guarantee-loss waterfall ODD robustness counterfactual model mechanism isolation

Core COMPLETED + Extended NOT_TRIGGERED/NOT_QUALIFIED " "

Step 0

- run inventory collect window trace collision actor history record Bridge/SCB
- run analysis directory
- run record run record

Step 1

- profile parse error channel
- timestamp dictionary clock alignment phase audit
- NOT_TESTABLE “ ”

Step 2 seed cause

- margin local minimum response-not-formed contact/collision L4/L3 diagnostic_seed
- seed physical conflict/control-demand change predicate t_c
- cause event target geometry effect predicate t_c discovery provenance
- / seed IDs cause-event IDs cause L4 Reaction verdict NOT_TESTABLE_NO_INDEPENDENT_CAUSE

loop catalog longitudinal lateral localization-feedback trajectory-tracking mode/context window Loop-Centric t_c

Step 3

- P_TARGET P_FUNC
- fallback/infeasibility
- Guardian Bridge

Step 4 paths

- DAG/config/trace/record channel dependency bundles source-to-effect paths join semantics input role
- path forward causal tracing backward provenance
- lineage grade

multi-input join source epoch pairwise skew raw/compensated coherence provenance “topic ” consumed inputs

Step 5 L3 observed chain metrics

- T_sampling=t_s-t_c
- T_software=t_cmd-t_s
- T_actuation=t_phys-t_cmd command-to-apply + apply-to-physical
- T_R=t_phys-t_c
- Age Vector publish/source-advance/fresh-effect gaps bundle coherence
- uncertainty

loop/window/mode/context sample interval/jitter feedback age/delay controller timing command gap/reuse actuation delay phase drift tracking/oscillation

Step 6 L2

- chain instance transport/wait/activation/exec/publish/reuse
- -run reference
- reference
- local symptom

execution/unresolved wait Core RESOURCE_EVIDENCE_REQUIRED Extended Resource Interference Graph wall duration CPU/GPU/lock

Step 7 L4-A L4-B contract

- / Architectural Contract scope metric semantics mode/context lock time
- physical safety model
- cut-off state dynamics/uncertainty
- R/A/G/C/L τ_R,τ_A,τ_G,τ_C,τ_L low/center/high
- anti-leakage domain qualification
- model/retrospective observed contract verdict

Step 8 observed correctness guarantee extension

- R/A/G/C/L architectural/physical uncertainty-aware verdict required components event/loop verdict
- effect-threshold sensitivity
- suffix/WCRT bounds slack waterfall guarantee-loss points
- formal bound observed waterfall single-run empirical headroom ASSURANCE_EXTENSION_NOT_ACTIVATED
- shrinking contract mode-transition/old-job/old-trajectory

Step 9 Backward diagnosis L1

- miss/loss seed segment
- hypothesis ledger equivalence classes defeaters discriminators
- resource candidate Resource Interference Graph
-

Step 10 L5/L6

- observed wall-clock velocity D_response
- deadline D_debt
- margin/outcome

- observed
- collision run right-censoring

Step 11

- clock phase target function initial state braking geometry freshness outcome conflict defeaters
- Evidence Ledger -> Claim Ledger -> Defeater Ledger
-

12.

12.1

- Perception target / tracking continuity
- Prediction target/trajectory
- Planning STOP/decision/fallback/infeasibility/latency debug
- Control command/latency
- Localization speed/acceleration/pose
- Bridge/SCB requested/applied delay
- collision and actor history

timestamp

12.2 Trace

e2e_trace_v3

```
events: event_id, trace_id, module, phase, mono_ns, pid, tid
message_context: edge, channel, input_seq, output_seq,
                 data_ts_ns, primary_parent_trace_id,
                 parent_trace_count, trace_valid
```

- Grade A/B lineage
- proc/runonce/substage timing
- source timestamp
- parent/fusion lineage
- overwrite/skip

monotonic trace trace anchor uncertainty

12.3 Parsed Cyber record

record

- record_message_index channel coverage/rate/gap
- Localization/Chassis state
- Planning header task latency decision trajectory
- Control command latency vehicle state Planning reuse/age
- Perception/Prediction obstacles source headers
- sensor timeline
- Planning-Control alignment sensor-to-Control message diagnostics

- channel protobuf
- record_time / task start
- sensor_to_control_reaction_time source-to-Control t_c t_phys physical Reaction Time
- obstacle table obstacle row gap channel timeline
- Planning total_time_ms task row frame statistic planning_seq

run record

- run record
- record-only Age/reuse/message payload
- trace/log
- L3 lineage L2 mechanism

12.4 Collision

- collision event/JSONL outcome
- actor history realtime factor
- Localization wall-time velocity
- CARLA frame/sim time e2e_sim_frames realtime_factor

13.

Core/Extended Core UNAVAILABLE + missing_reason Extended analysis_profile_status.csv

```
analysis/<run_id>/
├── report/
│   └── single_run_realtime_diagnosis.md
```

```
│   ├── tables/
│   │   ├── input_inventory.csv
│   │   ├── analysis_profile_status.csv
│   │   ├── event_catalog.csv
│   │   ├── event_timeline.csv
│   │   ├── loop_catalog.csv
│   │   ├── loop_windows.csv
│   │   ├── loop_timing_metrics.csv
│   │   ├── loop_physical_behavior.csv
│   │   ├── chain_catalog.csv
│   │   ├── chain_instances.csv
│   │   ├── dependency_bundles.csv
│   │   ├── partition_lineage.csv
│   │   ├── local_timing_primitives.csv
│   │   ├── age_vector.csv
│   │   ├── temporal_coherence.csv
│   │   ├── fresh_update_gaps.csv
│   │   ├── effect_predicate_registry.csv
│   │   ├── effect_threshold_sensitivity.csv
│   │   ├── temporal_fault_signature.csv
│   │   ├── clock_alignment_audit.csv
│   │   ├── phase_audit.csv
│   │   ├── target_identity_audit.csv
│   │   ├── functional_correctness_audit.csv
│   │   ├── requirement_registry.csv
│   │   ├── architectural_contracts.csv
│   │   ├── dynamic_contract_construction.csv
│   │   ├── contract_verdicts.csv
│   │   ├── velocity_trajectory_observed.csv
│   │   ├── physical_budget_observed.csv
│   │   ├── physical_outcomes.csv
│   │   ├── l5_recomputation.csv
│   │   ├── evidence_ledger.csv
│   │   ├── claim_ledger.csv
│   │   ├── claim_edges.csv
│   │   ├── claim_evidence_summary.csv
│   │   ├── exclusions_and_missing.csv
│   │   └── extended/
│   │       ├── slack_waterfall.csv
│   │       ├── guarantee_loss_points.csv
│   │       ├── timing_model_registry.csv
│   │       ├── path_and_suffix_bounds.csv
│   │       ├── bound_assumption_audit.csv
│   │       ├── resource_interference_graph.csv
│   │       ├── resource_interference_edges.csv
│   │       ├── physical_budget_model_predicted.csv
│   │       ├── diagnosis_hypothesis_ledger.csv
│   │       ├── diagnosis_edges.csv
│   │       ├── defeater_ledger.csv
│   │       └── counterfactual_model_audit.csv
└── validation/
    ├── data_quality_audit.md
    ├── anti_leakage_audit.md
    ├── arithmetic_recomputation.md
    ├── claim_audit.md
    └── validation.json
```

13.1

```
Event / Target / Geometry
Cause predicate and t_c interval
```

```
Selected critical chain(s) and lineage grade

Observed chain timing
  T_sampling / T_software / T_command_to_apply / T_actuation / T_R
  Age Vector / A_critical basis
  G_publish / G_source_advance / G_fresh_effect
  raw / compensated bundle coherence + pairwise source skew

Loop timing (when applicable)
  loop / window / mode / context
  T_s / J_s / feedback delay / controller timing / command hold-reuse / actuation
  physical tracking error / overshoot / oscillation

Architectural contracts
  R/A/G/C/L requirement IDs, provenance, scope and qualification
  per-component and composite observed verdict

Physical dynamic contracts
  tau_R low/center/high + qualification
  tau_A / tau_G / tau_C / tau_L if independently available
  per-component and composite observed verdict

Effect endpoint
  predicate spec and t_phys L/C/U
  threshold/hold/filter sensitivity and verdict stability

Assurance extension (only when activated)
  formal/analytical guarantee status and S_guar
  empirical headroom, explicitly non-guarantee

Slack waterfall
  per-segment observed budget use
  guarantee-not-established-from-start or first conditional proof-margin-negative boundary
  first irreversible observed miss
  dominant over-budget segment relative to declared reference

Diagnosis
  localized segment
  mechanism candidates and evidence
  resource interference subgraph when execution/wait is implicated
  observational equivalence class
  competing functional/clock/phase/physical explanations
  diagnosis strength and missing discriminator

Physical propagation
  D_response_wall_integral_data_observed_m
  D_debt_requirement_constrained_m
  minimum_clearance_data_observed_m / contact / collision
  safety_margin_requirement_constrained_derived_m
  Delta_M_phys_model_predicted_m
  right-censoring status

Outcome and allowed conclusion
```

13.2

analysis_profile_status.csv	run_id,core_status,extended_trigger,extended_status,qualification_gap,n

	otes
loop_catalog.csv	run_id,loop_id,controller,plant,feedback_sources,command_sink,sample_semantics,hold_semantics,source_evidence_ids
loop_windows.csv	loop_id,window_id,start_wall,end_wall,selection_rule,selected_before_outcome,mode_id,context_id,context_bounds,homogeneity_status,taint_tags
loop_timing_metrics.csv	loop_id,window_id,instance_id,metric_name,value,unit,source_event_id,effect_event_id,clock_domain,error_bound,reference_id,verdict,missing_reason
loop_physical_behavior.csv	loop_id,window_id,behavior_metric,value,unit,endpoint_definition,reference_id,verdict,lineage_to_timing_grade
temporal_coherence.csv	run_id,event_or_window_id,bundle_id,consumer_instance_id,input_id,input_role,source_time,source_semantics,read_time,raw_skew_ms,compensated_epoch,compensated_skew_ms,compensation_model,clock_error_ms,requirement_id,verdict,lineage_grade,missing_reason
effect_predicate_registry.csv	effect_spec_id,event_type,signal,comparison,threshold,hold_ms,hysteresis,filter,sample_basis,lock_time,provenance,qualification
effect_threshold_sensitivity.csv	event_id,effect_spec_id,threshold,hold_ms,filter,t_phys_low,t_phys_center,t_phys_high,T_R_low,T_R_center,T_R_high,contract_verdict,verdict_stability
architectural_contracts.csv	requirement_id,owner,artifact,version,scope_type,scope_id,dimension,metric_semantics,low,center,high,unit,mode_context,effective_time,lock_time,evidence_class,qualification
resource_interference_graph.csv	node_id,node_type,host,process,thread,resource,instance_scope,start,end,evidence_ids,availability
resource_interference_edges.csv	source_node,target_node,relation,overlap_ms,priority_or_blocking_detail,evidence_ids,status,residual_uncertainty

source_file/source_locator source_evidence_ids missing_reason reference_id/requirement_id requirement registry skew loop P_CLOCK error budget

14. Claim–Evidence

I-M-E-C-O-N

- I / Inputs
- M / Metrics

- E / Evidence /
- C / Criterion
- O / Output verdict confidence ceiling
- N / Next

P_CLOCK	anchor/sync offset/drift/residual resolution error budget
P_PHASE	producer/consumer/tick period phase origin phase sensitivity
P_TARGET	event/chain/outcome
P_FUNC	Bridge
P_BUNDLE	multi-input join role consumed instance source epoch compensation provenance
P_LOOP	loop plant/controller/feedback window selection mode/context
P_ARCH_CONTRACT	owner/artifact/version/scope/metric/unit/mode/context/lock time
P_PHYS_CONTRACT	outcome leakage uncertainty/domain audit
P_EFFECT	effect predicate outcome L/C/U threshold sensitivity

P_ARCH_CONTRACT P_PHYS_CONTRACT P_BUNDLE Coherence P_LOOP Loop case-level observation P_EFFECT t_phys C3_EVENT/C4_PHYS/C5

14.1

ID	
C1_STRESSOR_ENTRY	stressor/ / entry OR
C2	R/A/G/C/L reference
C3_EVENT	/ effect
C3_LOOP	loop window sampling/feedback/actuation mode/context
C4_ARCH	observed timing Architectural Temporal Contract
C4_PHYS	observed timing Physical Dynamic Contract
G1_GUARANTEE	Assurance Extension / C4_ARCH/C4_PHYS

ID	
	Core
E1_EMPIRICAL	single-run empirical headroom guarantee/causal claim
C5	C4_PHYS=CLEARLY_MISSED requirement-constrained physical budget consumption C4_ARCH G1 loss
C6_OUTCOME	contact/collision/clearance outcome margin degraded threshold/reference
C7	

C7

14.2

Level	
0	/
1	reference
2	event cause-effect propagation loop timing association
3A	architectural requirement
3P	physical requirement
3G	formal/analytical guarantee status
4	timing
5	timing-dominated safety-failure candidate
6	SUT-intrinsic temporal defect

/

14.3 defeaters

```
clock alignment uncertainty
phase/tick quantization
target mismatch
functional failure or fallback
initial unsafe state
pre-cause treatment-induced state divergence
braking capability / friction / grade / curvature mismatch
geometry or actor association conflict
record/trace window incomplete
outcome truncation
deadline leakage or model-domain mismatch
```

```
freshness/reuse alternative
cross-source temporal incoherence or compensation error
loop-mode/context mixing or post-selected window
effect predicate / threshold / hold-rule instability
resource-interference observational equivalence
external injected fault versus intrinsic Apollo defect
```

defeater OPEN/UNKNOWN UNCERTAIN/PARTIAL_PASS

15.

15.1

```
CHAIN_IDENTIFIED
SEGMENT_LOCALIZED
TASK_OR_EDGE_LOCALIZED
MECHANISM_SUPPORTED
ROOT_CAUSE_ISOLATED
NOT_LOCALIZABLE
```

“ task” /reference over-consumption

15.2

```
SAMPLING
TRANSPORT
QUEUE_BACKLOG
ACTIVATION_PHASE
EXECUTION
PUBLISH_BACKPRESSURE
STALE_DATA
REUSE_SAMPLE_HOLD
OVERWRITE_SKIP
DROP_REORDER
MODE_TRANSITION
BRIDGE_DELAY
ACTUATION
CLOCK_ARTIFACT
FUNCTIONAL_ALTERNATIVE
PHYSICAL_ALTERNATIVE
UNRESOLVED_MIXTURE
```

15.3

- architectural_contract_miss_time/window
- physical_contract_miss_time/window
- observed_contract_miss_time_interval deadline

- Assurance Extension guarantee_not_established_from_start
- Assurance Extension first_conditional_proof_margin_negative_boundary/time machine code FIRST_CONDITIONAL_GUARANTEE_LOSS
- first_irreversible_observed_miss_time
- contract_shrink_crossing_time
- formal bound extension summary ASSURANCE_EXTENSION_NOT_ACTIVATED observed contract

15.4

```
supported mechanism
+ causal segment
+ consumed budget relative to reference
+ evidence IDs
+ counter-evidence
+ unresolved equivalent candidates
+ missing discriminator
```

15.5

provenance

```
L_physical= (D_response^obs,D_debt^req,C_min^obs,M_safe^req\ derived,Δ M_phys^model)
```

16.

40. collision -> deadline miss
41. deadline miss -> collision fallback
42. output exists -> function correct
43. record_time difference -> execution time record task start/end
44. Control publishes at 100 Hz -> fresh control every 10 ms Planning
45. MRT maximum = MDA maximum -> every RT equals every Age warm-up/chain
46. p-partition point -> root cause proven /
47. single-run P99/max -> formal guarantee run
48. first guarantee-loss point -> fault origin
49. model deadline -> observed deadline miss
50. same-run stopping distance -> independent deadline
51. CARLA sim frame displacement -> D_response

52. collision-truncated trajectory -> full observed stopping distance
53. Bridge injection -> Apollo intrinsic defect
54. D_response + D_debt both subtracted debt response distance
55. diagnostic seed -> physical cause/t_c response-not-formed margin minimum contact/collision cause
56. guarantee loss -> C5 physical loss observed contract miss timing-caused physical budget loss
57. one input path -> whole fusion cause dependency-bundle completeness path
58. clearance - safety threshold -> pure observed margin requirement-constrained derived model/predicted
59. every input age within limit -> bundle coherent source-time skew
60. same header/fusion epoch -> temporal coherence measurement time residual uncertainty
61. single loop jitter spike -> closed-loop instability mode/context / margin
62. architectural miss -> physical safety miss
63. no physical deadline -> no real-time problem Architectural Contract
64. wall execution tail -> CPU contention Resource Interference Graph /
65. choose effect threshold after outcome -> stable t_phys predicate
66. missing WCRT -> observed diagnosis incomplete Assurance Extension

---

17. run

event	ground truth/target/ predicate	EVENT_UNCERTAIN
chain	trace/provenance/headers	ASSOCIATION_ONLY / NOT_TESTABLE
	read/start/end/write	UNRESOLVED_SEGMENT
	consumer instance source epochs + qualified coherence reference	COHERENCE_OBSERVED / NOT_TESTABLE
	loop/window/mode/context + sampling/feedback/actuation + reference	LOOP_OBSERVATION_ONLY / NOT_TESTABLE
	observed compatible metric + qualified architectural contract	ARCH_REQUIREMENT_MISSING / NOT_TESTABLE
	observed timing + qualified physical deadline/margin	MODEL_ONLY / RETROSPECTIVE_ONLY / NOT_TESTABLE

	qualified path/suffix upper bounds Extended	ASSURANCE_EXTENSION_NOT_ACTIVATED
	+ alternatives	CANDIDATE_SET_ONLY
	wall velocity + endpoints + qualified deadline	RESPONSE_DISTANCE_ONLY / UNAVAILABLE
	collision/geometry/complete trajectory	OUTCOME_UNCERTAIN
Apollo		NOT_ESTABLISHED_FROM_SINGLE_RUN

NOT_TESTABLE run

18.

- event predicate t_c
- target
- chain causal instance
- forward backward lineage warm-up/overwrite

- Reaction Age Gap Coherence Loop
- Coherence consumer bundle topic nearest row
- Loop window mode/context outlier
- software command physical effect
- record/header/source/trace/wall/sim time
- clock error slack verdict

Contract

- Architectural Physical Contract scope verdict
- Architectural threshold artifact run
- physical deadline
- cut-off run outcome
- geometry/model/domain

- low/center/high
- observed correctness formal guarantee empirical headroom
- Effect predicate L/C/U sensitivity

- guarantee-loss point root cause
- overrun reference/budget
- phase/queue/execution discriminator
- execution Resource Interference Graph residual
- backward hypothesis forward proof

- D_response
- D_debt deadline
- moving target closest approach
- observed model
- collision truncation

- defeaters
- claim level
-

19.

19.1 ECRTS 2023

cause-effect chain read/write job immediate forward/backward chain MRT/MDA partition warm-up path 3 MRT/MDA job-chain 4 p-partitioned job chains 5 Theorem 14 MRT/MDA 6

source/sink DAG source-to-sink path dependency-bundle coherence AND-join theorem

MRT=MDA Reaction=Age partition theorem root-cause localization theorem path timing multi-input bundle coherence job read/write

19.2 Yi et al. 2021

sensor arrival E2E deadline source-to-sink path constraint task schedulability E2E guarantee shrinking deadline/mode-transition Architectural Contract Physical Dynamic Contract Observed Core Formal Assurance Extension II E2E deadline III Eq. (16) 2Σ p_i IV relaxing/shrinking deadline mode-change protocol

CPU EDF WCET 2Σ p_i Apollo/Cyber RT bound — deadline

19.3 Koopman, Osyk & Weast 2019

context-dependent safety envelope response time braking trajectory closest approach friction/slope/curvature/vehicle/environment variability ODD 3.1 RSS following-distance/response-time 4 Eqs. (2)–(4) closest-approach edge cases 5.1–5.2 μODD

RSS-like ODD /

19.4

run

- Event-Centric/Loop-Centric
- [R,A,G,C,L] bundle coherence
- Architectural/Physical contract Core/Extended profiles
- effect-threshold sensitivity Resource Interference Graph
- tau_robust(m) context-region
- slack derivative dot S T_exhaust_local
- conditional guarantee-loss waterfall
- requirement-constrained distance debt Claim–Evidence

/ evidence gates

20.

— — — — —

$$\{ (U_E,U_L) \longrightarrow (R,A,G,C,L)^{obs} \parallel (\tau_{arch}^{req},\tau_{phys}^{req}(x)) \}$$

Architectural Contract Physical Dynamic Contract event chain loop window

{ S_arch=τ_arch^req- T^obs, S_phys=τ_phys^req(x)- T^obs }

Assurance Extension

{S_guar=τ_R,low^req-R_path^UB}

Core / observed miss coherence/loop anomalies Extended chain guarantee slack waterfall " " miss L2 primitives Resource Interference Graph L1

{ D_response^obs, D_debt^req, C_min^obs, M_safe^req\ derived, Δ M_phys^model, Outcome^obs }

> 300 ms

> U_E U_L observed Reaction Freshness Continuity Cross-Source Coherence Loop Timing Assurance Extension
> requirement-constrained

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